

Caoyuan Ma

University of Tokyo
Graduate School of Information Science and Technology
Ph.D. Student — Multimodal LLM Safety

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Education

The University of Tokyo	<i>Oct. 2025 – Present</i>
Ph.D. in Information Science and Technology	Research Focus: Multimodal LLM Safety
Wuhan University, School of Computer Science	<i>Sep. 2022 – Jun. 2025</i>
M.S. in Computer Science and Technology	Human-centric AIGC and 3D Vision
Wuhan University, School of Computer Science	<i>Sep. 2018 – Jul. 2022</i>
B.Eng. in Software Engineering	GPA: 3.87/4.00; top 1% in two academic years

Research Interests

Research Interests: Multimodal intelligence, with a focus on post-training, alignment, safety, and embodied spatial reasoning. [🔗 Research portfolio](#)

Research and Industry Experience

Multimodal Foundation Models

JD Explore Academy — Research Intern	<i>Apr. 2026 – Present</i>
– Audited every sample across six benchmarks, categorized image and question patterns, and built targeted collection, cleaning, and multi-agent cross-validation pipelines that produced 7.74 million samples totaling 15.6 billion tokens. – Improved the average score of Qwen3-VL-30B-A3B from 48.3 to 55.5; optimized the data mixture for the mainline model, enabling the specialist model to match top-tier 300B-model performance. – Built data for ZeroBench-main, improving the specialist model by approximately 10 points during SFT; developed CoT data collection pipelines for Claude Opus and Fable to support agent training.	

Shanghai AI Laboratory (Remote) — Research Intern	<i>Sep. 2025 – Apr. 2026</i>
– Supported base-model experiments and evaluation for <i>SafeWork-R1</i>. – Independently studied semi-self-evolving LLM safety; manuscript in revision.	

Video Generation

Hedra (U.S., Remote) — Research Intern	<i>2025</i>
– Selected and applied DMD and Self-Forcing to production video models, delaying noticeable visual degradation from approximately 8 to 30 seconds.	

StepFun — Research Intern	<i>2025</i>
– Led a four-person team to build a large-scale video-data curation pipeline for model training, processing more than 30 TB of video.	

3D Human Reconstruction

Noitom & JD Group — Research Intern	<i>2022 – 2024</i>
– Built a synchronized IMU motion-capture and multi-camera system to collect multi-view human motion data. – Developed a NeRF-based human-centric 6D reconstruction method spanning 3D space and 3D skeleton-driven motion, resulting in the first-authored CVPR 2024 paper <i>HumanNeRF-SE</i>.	

Awards and Honors

- **Scholarships:** National Scholarship (twice; 2/208), First-Class Scholarship (twice; top 5%), and SPRING GX.
- **Honor:** Outstanding Graduate of Wuhan University (top 10%).
- **Competitions:** Meritorious Winner, Mathematical Contest in Modeling; First Prize, National Undergraduate IoT Contest.

Publications

*Equal contribution. My name is shown in **bold**. Conference/journal versions supersede duplicate preprints.

1. **HumanNeRF-SE: A Simple yet Effective Approach to Animate HumanNeRF with Diverse Poses**
Caoyuan Ma, Yu-Lun Liu, Zhixiang Wang, Wu Liu, Xincheng Liu, and Zheng Wang. *CVPR* 2024.
2. **Learning a Dynamic Neural Human via Poses Guided Dual Spaces Feature**
Caoyuan Ma, Runqi Wang, Wu Liu, Zhiming Zhou, and Zheng Wang. *IEEE AVSS* 2024.
3. **SafeCap: Improving LVLm Safety with Image Captioning Reinforcement Learning**
Caoyuan Ma, Liuwenpu, Weichu Xie, Tian Gu, Shilei Zhao, Lingxi Min, Shuai Dong, Yuqi Xu, Ji Zhao, Ziyue Wang, et al.. *arXiv:2608.10513*, 2026.
4. **SafeRI: Recognition and Intervention for Token-Level Safety Intervention in Large Vision Language Models**
Caoyuan Ma*, Tian Gu*, Liuwenpu, Weichu Xie, Shuai Dong, Yuqi Xu, Ji Zhao, Ziyue Wang, et al.. *arXiv:2609.03544*, 2026.
5. **You Think, You Act: The New Task of Arbitrary Text to Motion Generation**
Runqi Wang*, Caoyuan Ma*, Guopeng Li*, Hanrui Xu, Yizhou Li, and Zheng Wang. *ICCV* 2025.
6. **Subjective Camera 1.0: Bridging Human Cognition and Visual Reconstruction through Sequence-Aware Sketch-Guided Diffusion**
Haoyang Chen*, Dongfang Sun*, Caoyuan Ma*, Shiqin Wang, Kewei Zhang, Zheng Wang, and Zhixiang Wang. *ICCV* 2025.
7. **Leader is Guided: Interactive Motion Generation via Lead-Follow Paradigm and Trajectory Guidance**
Runqi Wang*, Caoyuan Ma*, Jian Zhao, Hanrui Xu, Dongfang Sun, Haoyang Chen, Lin Xiong, Zheng Wang, and Xuelong Li. *ACM MM* 2025.
8. **Behave Your Motion: Habit-Preserved Cross-Category Animal Motion Transfer**
Zhimin Zhang*, Bi'an Du*, Caoyuan Ma, Zheng Wang, and Wei Hu. *ACM MM* 2025.
9. **Causal Deciphering and Inpainting in Spatio-Temporal Dynamics via Diffusion Model**
Yifan Duan, Jian Zhao, Junyuan Mao, Hao Wu, Jingyu Xu, Shiqin Wang, Caoyuan Ma, Kai Wang, Kun Wang, and Xuelong Li. *NeurIPS* 2024.
10. **Improving Stack Overflow Question Title Generation with Copying Enhanced CodeBERT Model and Bi-Modal Information**
Fengji Zhang, Xiao Yu, Jacky Keung, Fuyang Li, Zhiwen Xie, Zhen Yang, Caoyuan Ma, and Zhimin Zhang. *Information and Software Technology*, 148:106922, 2022.
11. **Right Makes Might: Aligning Verified Hidden States Empowers RL Reasoning**
Ziyue Wang, Aolin Yuan, Yongfu Zhu, Shuai Dong, Wu Liu, et al., including Caoyuan Ma. *arXiv:2606.03234*, 2026.
12. **SafeWork-R1: Coevolving Safety and Intelligence under the AI-45 Law**
Shanghai AI Laboratory et al., including Caoyuan Ma. *arXiv:2507.18576*, 2025.